

QUADRATIC EQUATION & EXPRESSIONS

Syllabus

Theory of quadratic equations and quadratic expressions, relation between the roots and coefficients, sign of a quadratic expression, greatest and least values of a quadratic expression.

Quadratic Equation

An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$ and a, b, c are real numbers, is called a quadratic equation. The numbers a, b, c are called the coefficients of the quadratic equation. A root of the quadratic equation is a number α (real or complex) such that $a\alpha^2 + b\alpha + c = 0$.

The roots of the quadratic equation are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Basic Results:

- ☐ The quantity D ($D = b^2 - 4ac$) is known as the discriminant of the quadratic equation.
- ☐ The quadratic equation has real and equal roots if and only if $D = 0$ i.e. $b^2 - 4ac = 0$.
- ☐ The quadratic equation has real and distinct roots if and only if $D > 0$ i.e. $b^2 - 4ac > 0$.
- ☐ The quadratic equation has complex roots with non-zero imaginary parts if and only if $D < 0$ i.e. $b^2 - 4ac < 0$.
- ☐ If $p + iq$ (p and q being real) is a root of the quadratic equation where $i = \sqrt{-1}$, then $p - iq$ is also a root of the quadratic equation.
- ☐ If $p + \sqrt{q}$ is an irrational root of the quadratic equation, then $p - \sqrt{q}$ is also a root of the quadratic equation provided that all the coefficients are rational.
- ☐ The quadratic equation has rational roots if D is a perfect square and a, b, c are rational. If $a = 1$ and b, c are integers and the roots of the quadratic equation are rational, then the roots must be integers.
- ☐ If the quadratic equation is satisfied by more than two numbers (real or complex), then it becomes an identity i.e. $a = b = c = 0$.
- ☐ Let α and β be two roots of the given quadratic equation. Then $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$.
A quadratic equation, whose roots are α and β can be written as $(x - \alpha)(x - \beta) = 0$
i.e., $ax^2 + bx + c \equiv a(x - \alpha)(x - \beta)$.

Illustration 1: Prove that the roots of the quadratic equation $ax^2 - 3bx - 4a = 0$ are real and distinct for all real a and b .

Solution: $D = (-3b)^2 - 4(-4a)(a) = 9b^2 + 16a^2$ which is always positive.
Hence the roots will be real and distinct.

Illustration 2: Find the value of m for which the equation $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has equal roots.

Solution: Given equation is $(1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$.

If roots are equal, then discriminant $= 4(1 + 3m)^2 - 4(1 + m)(1 + 8m) = 0$
 $\Rightarrow m^2 - 3m = 0$ or $m = 0, 3$.

Problems based on value of symmetrical functions of α and β , where α and β are roots of a quadratic equation.

Illustration 3: If α and β be roots of equation $ax^2 + bx + c = 0$, prove that $a\left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}\right) + b\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) = b$.

Solution:

Here $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$.

$$\begin{aligned} \text{Now, } a\left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}\right) + b\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) &= \frac{a(\alpha^3 + \beta^3) + b(\alpha^2 + \beta^2)}{\alpha\beta} \\ &= \frac{a[(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)] + b[(\alpha + \beta)^2 - 2\alpha\beta]}{\alpha\beta} \\ &= \frac{a\left[\left(-\frac{b}{a}\right)^3 - 3\frac{c}{a}\left(-\frac{b}{a}\right)\right] + b\left[\left(-\frac{b}{a}\right)^2 - 2\frac{c}{a}\right]}{c/a} = \frac{bc}{c/a} = b. \end{aligned}$$

Condition for two Quadratic Equations to have a Common Root

Let $ax^2 + bx + c = 0$ and $dx^2 + ex + f = 0$ have a common root α (say). Then $a\alpha^2 + b\alpha + c = 0$ and $d\alpha^2 + e\alpha + f = 0$.

Solving for α^2 and α , we get $\frac{\alpha^2}{bf - ce} = \frac{\alpha}{dc - af} = \frac{1}{ae - bd}$

$$\text{i.e. } \alpha^2 = \frac{bf - ce}{ae - bd} \text{ and } \alpha = \frac{dc - af}{ae - bd} \Rightarrow (dc - af)^2 = (bf - ce)(ae - bd),$$

which is the required condition for the two equations to have a common root.

Note: For both roots common, we have $\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$

Illustration 4: Find the value of p if the equations $3x^2 - 2x + p = 0$ and $6x^2 - 17x + 12 = 0$ have a common root.

Solution:

Let α be the common root, then $3\alpha^2 - 2\alpha + p = 0$,(1)

and $6\alpha^2 - 17\alpha + 12 = 0$ (2)

$$\Rightarrow \frac{\alpha^2}{-24 + 17p} = \frac{\alpha}{6p - 36} = \frac{1}{-51 + 12} = -\frac{1}{39} \Rightarrow \alpha^2 = \frac{24 - 17p}{39}$$

$$\text{and } \alpha = \frac{36 - 6p}{39} \Rightarrow \frac{24 - 17p}{39} = \left(\frac{36 - 6p}{39}\right)^2 = \left(\frac{12 - 2p}{13}\right)^2$$

$$= \frac{144 - 48p + 4p^2}{169} \Rightarrow 312 - 221p = 432 - 144p + 12p^2$$

$$\Rightarrow 12p^2 + 77p + 120 = 0 \Rightarrow (4p + 15)(3p + 8) = 0 \Rightarrow p = -\frac{15}{4}, -\frac{8}{3}.$$

SOLVED PROBLEMS

SUBJECTIVE

Problem 1: Find the values of 'a' for which the equation $(x^2 + x + 2)^2 - (a - 3)(x^2 + x + 2)(x^2 + x + 1) + (a - 4)(x^2 + x + 1)^2 = 0$ has at least one real root.

Solution: The given equation can be rewritten as

$$\left(\frac{x^2 + x + 2}{x^2 + x + 1}\right)^2 - (a - 3)\left(\frac{x^2 + x + 2}{x^2 + x + 1}\right) + (a - 4) = 0.$$

Let $\frac{x^2 + x + 2}{x^2 + x + 1} = t$ or, $t = 1 + \frac{1}{x^2 + x + 1}$.

Since $x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$,

$$x^2 + x + 1 \geq \frac{3}{4} \Rightarrow t \in \left(1, \frac{7}{3}\right].$$

Now the given equation reduces to $t^2 - (a - 3)t + (a - 4) = 0$.

At least one root of this equation must lie in $\left(1, \frac{7}{3}\right]$.

Now, $t = \frac{(a - 3) \pm \sqrt{(a - 3)^2 - 4(a - 4)}}{2} \Rightarrow t = a - 4, 1$.

For one root to lie in $\left(1, \frac{7}{3}\right]$ we must have

$$1 < a - 4 \leq \frac{7}{3} \Rightarrow 5 < a \leq \frac{19}{3}.$$

Alternative Solution:

Putting $x^2 + x + 1 = \alpha$, we get

$$(\alpha + 1)^2 - (a - 3)\alpha(\alpha + 1) + (a - 4)\alpha^2 = 0$$

$$\Rightarrow \alpha(5 - a) + 1 = 0 \Rightarrow \alpha = -\frac{1}{5 - a} \Rightarrow x^2 + x + 1 = \frac{1}{a - 5}$$

$$\Rightarrow \frac{1}{a - 5} \geq 3/4 \Rightarrow a - 5 > 0 \text{ and } a - 5 \leq 4/3 \Rightarrow 5 < a \leq 19/3.$$

Problem 2: If α, β are the roots of $x^2 + px + q = 0$, and also of $x^{2n} + p^n x^n + q^n = 0$, and if α/β is a root of $x^n + 1 + (x + 1)^n = 0$, then prove that n must be an even integer, where $\alpha^n \neq \beta^n$.

Solution: α, β are roots of $x^2 + px + q = 0 \Rightarrow \alpha + \beta = -p$ (1)

Also α, β are roots of $x^{2n} + p^n x^n + q^n = 0$

$$\Rightarrow \alpha^{2n} + p^n \alpha^n + q^n = 0, \quad \dots (2)$$

$$\beta^{2n} + p^n \beta^n + q^n = 0. \quad \dots (3)$$

Subtract (3) from (2). We get $\alpha^{2n} - \beta^{2n} + p^n(\alpha^n - \beta^n) = 0$

$$\Rightarrow \alpha^n + \beta^n = -p^n. \quad (\text{as } \alpha^n \neq \beta^n) \quad \dots (4)$$

Now $\frac{\alpha}{\beta}$ is a root of $x^n + 1 + (x + 1)^n = 0 \Rightarrow \left(\frac{\alpha}{\beta}\right)^n + 1 + (\alpha/\beta + 1)^n = 0$

$$\Rightarrow \alpha^n + \beta^n + (\alpha + \beta)^n = 0 \Rightarrow -p^n + (-p)^n = 0$$

from (1) and (4),

$$\Rightarrow -p^n + (-1)^n p^n = 0. \text{ This is true only if } n \text{ is an even integer.}$$

OBJECTIVE

- Problem 1:** If α and β are the roots of the equation $2x^2 - 3x - 6 = 0$, then equation whose roots are $\alpha^2 + 2, \beta^2 + 2$, is
- (A) $4x^2 + 49x + 118 = 0$ (B) $4x^2 - 49x + 118 = 0$
 (C) $4x^2 - 49x - 118 = 0$ (D) $x^2 - 49x + 118 = 0$

Solution: Here $\alpha + \beta = 3/2, \alpha\beta = -6/2 = -3$ so that

$$S = \alpha^2 + \beta^2 + 4 = (\alpha + \beta)^2 - 2\alpha\beta + 4 = \frac{49}{4},$$

$$P = \alpha^2\beta^2 + 2(\alpha^2 + \beta^2) + 4 = \alpha^2\beta^2 + 4 + 2[(\alpha + \beta)^2 - 2\alpha\beta] = \frac{118}{4}.$$

$$\text{Therefore, the equation is } x^2 - \left(\frac{49}{4}\right)x + \left(\frac{118}{4}\right) = 0$$

$$\Rightarrow 4x^2 - 49x + 118 = 0.$$

Hence (B) is the correct answer.

Alternate: Let $y = x^2 + 2$, then $2x^2 - 3x - 6 = 0$

$$\Rightarrow (3x)^2 = (2x^2 - 6)^2 \Rightarrow [2(y - 2) - 6]^2 = 9(y - 2) \Rightarrow 4y^2 - 49y + 118 = 0.$$

- Problem 2:** If the roots of the equation $x^2 - px + q = 0$ differ by unity then
- (A) $p^2 = 1 - 4q$ (B) $p^2 = 1 + 4q$
 (C) $q^2 = 1 - 4p$ (D) $q^2 = 1 + 4p$

Solution: Suppose the equation $x^2 - px + q = 0$ has the roots $\alpha + 1$ and α then $\alpha + 1 + \alpha = p \Rightarrow 2\alpha = p - 1 \dots (1)$
 and $(\alpha + 1)\alpha = q \Rightarrow \alpha^2 + \alpha = q. \dots (2)$

Putting the value of α from (1) in (2), we get

$$\frac{(p-1)^2}{4} + \frac{p-1}{2} = q \Rightarrow (p-1)^2 + 2(p-1) = 4q$$

$$\Rightarrow p^2 - 1 = 4q \Rightarrow p^2 = 4q + 1.$$

Alternative: Let α and β be the roots. $|\alpha - \beta| = 1 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 1$

$$\Rightarrow p^2 - 4q = 1, \text{ or } p^2 = 1 + 4q.$$

Hence (B) is the correct answer.

- Problem 3:** If p and q are the roots of the equation $x^2 + px + q = 0$, then
- (A) $p = 1, q = -2$ (B) $p = 0, q = 1$
 (C) $p = -2, q = 0$ (D) $p = -2, q = 1$

Solution: Since p and q are roots of the equation $x^2 + px + q = 0$,
 $p + q = -p$ and $pq = q \Rightarrow pq = q \Rightarrow q = 0$ or $p = 1$.
 If $q = 0$, then $p = 0$ and if $p = 1$, then $q = -2$.
 Hence (A) is the correct answer.

- Problem 4:** If $a, b \in \mathbb{R}$, $a \neq 0$ and the quadratic equation $ax^2 - bx + 1 = 0$ has imaginary roots, then $a + b + 1$ is
- (A) positive (B) negative
(C) zero (D) depends on the sign of b

Solution: Let $f(x) = ax^2 - bx + 1$.
Now $f(0) = 1$ and roots are imaginary, $f(x) > 0 \forall x \in \mathbb{R}$
 $\Rightarrow f(-1) = a + b + 1 > 0$.
Hence (A) is the correct answer.

- Problem 5:** (α, β) ; (β, γ) and (γ, α) are respectively the roots of $x^2 - 2px + 2 = 0$, $x^2 - 2qx + 3 = 0$ and $x^2 - 2rx + 6 = 0$. If α, β and γ are all positive, then the value of $p + q + r$ is
- (A) 2 (B) 3
(C) 6 (D) 11

Solution: We have, $\alpha\beta = 2$, $\beta\gamma = 3$ and $\gamma\alpha = 6$,
then $\alpha\beta \cdot \beta\gamma \cdot \gamma\alpha = 2 \cdot 3 \cdot 6 \Rightarrow \alpha\beta\gamma = 6$ and α, β, γ are all positive.
Thus we have $\alpha = 2$, $\beta = 1$, $\gamma = 3$
Also, $2p = \alpha + \beta = 3$, $2q = \beta + \gamma = 4$ and $2r = \gamma + \alpha = 5$.
Hence $p + q + r = 6$.
Hence (C) is the correct answer.

- Problem 6:** The number of solution(s) of the equation $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$, is equal to
- (A) 2 (B) 4
(C) no solution (D) infinitely many solutions

Solution: We have, $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$
 $\Rightarrow |\sqrt{x} - 2| + (\sqrt{x})^2 - 4\sqrt{x} + 2 = 0$
 $\Rightarrow |\sqrt{x} - 2| + (\sqrt{x} - 2)^2 - 2 = 0 \Rightarrow |\sqrt{x} - 2| + |\sqrt{x} - 2|^2 - 2 = 0$
 $\Rightarrow |\sqrt{x} - 2| = -2, 1$ [-2 is unacceptable]
Hence $\sqrt{x} - 2 = \pm 1$ or $x = 1, 9$.
Hence (A) is the correct answer.

- Problem 7:** If $x^2 + ax + b$ is an integer for every integer x then
- (A) 'a' is always an integer but 'b' need not be an integer
(B) 'b' is always an integer but 'a' need not be an integer
(C) a and b are always integers.
(D) none of these

Solution: Let $f(x) = x^2 + ax + b$
Clearly, $f(0) = b \Rightarrow b$ is an integer.
Now $f(1) = 1 + a + b \Rightarrow a$ is an integer.
Hence (C) is the correct answer.

- Problem 8:** If $x^2 - 4x + \log_{1/2} a = 0$ does not have two distinct real roots, then maximum value of a is
- (A) $1/4$ (B) $1/16$
(C) $-1/4$ (D) none of these

Solution: Since $x^2 - 4x + \log_{1/2} a = 0$ does not have two distinct real roots,
discriminant ≤ 0
 $\Rightarrow 16 - 4 \log_{1/2} a \leq 0$
 $\Rightarrow \log_{1/2} a \geq 4 \Rightarrow a \leq 1/16$.
Hence (B) is the correct answer.

Problem 9: Sum of the real roots of the equation $x^2 + 5|x| + 6 = 0$
(A) equals to 5 (B) equals to 10
(C) equals to -5 (D) does not exist

Solution: Since x^2 , $5|x|$ and 6 are positive so $x^2 + 5|x| + 6 = 0$ does not have any real root.
Therefore sum does not exist.
Hence (D) is the correct answer.

EXERCISE – 1

- Show that if the roots of the equation $(a^2 + b^2)x^2 + 2x(ac + bd) + c^2 + d^2 = 0$ are real, they will be equal.
- Find for what value of 'a' equation $x^2 + 2x - (a^3 - 3a - 3) = 0$ has real roots.
- Prove that the roots of the equation $bx^2 + (b - c)x + (b - c - a) = 0$ are real if those of $ax^2 + 2bx + b = 0$ are imaginary.
- If α, β are the roots of the equation $x^2 - px + q = 0$ and α_1, β_1 be the roots of the equation $x^2 - qx + p = 0$, then form the quadratic equation whose roots are $\frac{1}{\alpha_1\beta} + \frac{1}{\alpha\beta_1}$ and $\frac{1}{\alpha\alpha_1} + \frac{1}{\beta\beta_1}$.
- If two equations $x^2 + cx + ab = 0$ and $x^2 + bx + ca = 0$ have a common root, then show that $a + b + c = 0$.
- Prove that equations $(q - r)x^2 + (r - p)x + p - q = 0$ and $(r - p)x^2 + (p - q)x + q - r = 0$ have a common root.
- If the equations $ax^2 + bx + c = 0$ and $x^2 + 2x + 4 = 0$ have a common root then find a: b: c.
- If the equations $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have a common root and $a, b, c \in \mathbb{N}$ then find the minimum value of $a + b + c$.
- If $b \in \mathbb{R}^+$ then the roots of the equation $(2 + b)x^2 + (3 + b)x + (4 + b) = 0$ is
(A) real and distinct (B) real and equal
(C) imaginary (D) cannot predicted
- If the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have exactly one common root, then the numerical value of $a + b$ is
(A) 1 (B) -1
(C) 0 (D) none of these

Relation between the Roots of a Polynomial Equation of Degree n

Consider the equation

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 = 0 \quad \dots (1)$$

($a_0, a_1, \dots, a_n$ are real coefficients and $a_n \neq 0$).

Let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of equation (1). Then

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 \equiv a_n (x - \alpha_1) (x - \alpha_2) \dots (x - \alpha_n).$$

Comparing the coefficients of like powers of x , we get

$$\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_n = -\frac{a_{n-1}}{a_n},$$

$$\alpha_1 \alpha_2 + \alpha_1 \alpha_3 + \alpha_1 \alpha_4 + \dots + \alpha_2 \alpha_3 + \dots + \alpha_{n-1} \alpha_n = \frac{a_{n-2}}{a_n},$$

$$\alpha_1 \alpha_2 \dots \alpha_r + \dots + \alpha_{n-r+1} \alpha_{n-r+2} \dots \alpha_n = (-1)^r \frac{a_{n-r}}{a_n},$$

$$\alpha_1 \alpha_2 \dots \alpha_n = (-1)^n \frac{a_0}{a_n}.$$

e.g. If α, β, γ and δ are the roots of $ax^4 + bx^3 + cx^2 + dx + e = 0$ then

$$\alpha + \beta + \gamma + \delta = -b/a,$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = c/a,$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -d/a,$$

$$\alpha\beta\gamma\delta = e/a.$$

Notes:

- (i) A polynomial equation of degree n has n roots (real or imaginary).
- (ii) If all the coefficients are real then the imaginary roots occur in pairs i.e. number of complex roots is always even.
- (iii) If the degree of a polynomial equation is odd then the number of real roots will also be odd. It follows that at least one of the roots will be real.
- (iv) If α is repeated root (repeating r times) of a polynomial equation $f(x) = 0$ of degree n i.e. $f(x) = (x - \alpha)^r g(x)$, where $g(x)$ is a polynomial of degree $n - r$ and $g(\alpha) \neq 0$, then $f(\alpha) = f'(\alpha) = f''(\alpha) = \dots = f^{(r-1)}(\alpha) = 0$ and $f^{(r)}(\alpha) \neq 0$.
- (v) **Remainder Theorem:** If we divide a polynomial $p(x)$ by $x - \alpha$, the remainder obtained is $p(\alpha)$. Note that if $p(\alpha) = 0$, then $x - \alpha$ is a factor of $p(x)$.
- (vi) If a polynomial equation of degree n has $n + 1$ roots say $x_1, \dots, x_{n+1}$, ($x_i \neq x_j$ if $i \neq j$), then the polynomial is identically zero. i.e. $p(x) = 0, \forall x \in \mathbb{R}$.
(In other words, the coefficients $a_0, \dots, a_n$ are all zero).
- (vii) If $p(a)$ and $p(b)$ ($a < b$) are of opposite sign, then $p(x) = 0$ has odd number of roots in (a, b) , i.e. it has at least one root in (a, b) .
- (viii) If coefficients in $p(x)$ have ' m ' changes in signs, then $p(x) = 0$ have at most ' m ' positive real roots and if $p(-x)$ have ' t ' changes in sign, then $p(x) = 0$ have at most ' t ' negative real roots. By this we can find maximum number of real roots and minimum number of complex roots of a polynomial equations $p(x) = 0$.

Illustration 1: Let a and b be two roots of the equation $x^3 + px^2 + qx + r = 0$ satisfying the relation $ab + 1 = 0$. Prove that $r^2 + pr + q + 1 = 0$. ($r \neq 0$)

Solution:

Given equation is $x^3 + px^2 + qx + r = 0$.

Let the third root be c , so that $abc = -r$(1)

Given $ab = -1$

$\Rightarrow$ from (1) $c = r$, which is a root of the given equation.

$$\Rightarrow (r)^3 + p(r)^2 + q(r) + r = 0 \Rightarrow r^2 + pr + q + 1 = 0.$$

Illustration 2: If $b^2 < 2ac$, then prove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

Solution:

Let α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$.

$$\text{Then } \alpha + \beta + \gamma = -\frac{b}{a}, \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \alpha\beta\gamma = \frac{-d}{a}$$

$$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$, which is not possible if all α, β, γ are real. So at least one root is non-real, but complex roots occur in pairs. Hence given cubic equation has two non-real and one real roots.

Alternative Solution:

Let $f(x) = ax^3 + bx^2 + cx + d$ and $f(x) = 0$ have all the roots real

$\Rightarrow f'(x) = 3ax^2 + 2bx + c = 0$ has two real roots.

But its discriminant

$$= (2b)^2 - 4 \cdot 3 \cdot ac = (b^2 - 2ac) - ac < 0 \quad (\text{as } b^2 < 2ac)$$

which is a contradiction $\Rightarrow f(x) = 0$ will not have all the roots real.

The Method of Intervals (Wavy Curve Method)

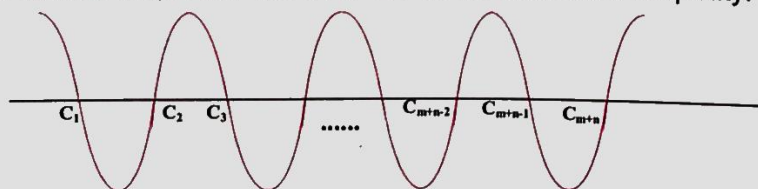
The Method of intervals (or wavy curve) is used for solving inequalities of the form

$$f(x) = \frac{(x - a_1)^{n_1} (x - a_2)^{n_2} \dots (x - a_k)^{n_k}}{(x - b_1)^{m_1} (x - b_2)^{m_2} \dots (x - b_p)^{m_p}} > 0 \quad (< 0, \leq 0, \text{ or } \geq 0) \text{ where } n_1, n_2, \dots, n_k, m_1, m_2, \dots, m_p \text{ are}$$

natural numbers and the numbers $a_1, a_2, \dots, a_k; b_1, b_2, \dots, b_p$ are any real numbers such that $a_i \neq b_j$, where $i = 1, 2, 3, \dots, k$ and $j = 1, 2, 3, \dots, p$.

It consists of the following statements:

- ☐ All zeros¹ of the function $f(x)$ contained on the left hand side of the inequality should be marked on the number line with inked (black) circles.
- ☐ All points of discontinuities² of the function $f(x)$ contained on the left hand side of the inequality should be marked on the number line with un-inked (white) circles.
- ☐ Check the value of $f(x)$ for any real number greater than the right most marked number on the number line.
- ☐ From right to left, beginning above the number line (in case of value of $f(x)$ is positive in step (iii), otherwise, from below the number line), a wavy curve should be drawn to pass through all the marked points so that when it passes through a simple point³, the curve intersects the number line, and, when passing through a double point⁴, the curve remains located on one side of the number line.
- ☐ The appropriate intervals are chosen in accordance with the sign of inequality (the function $f(x)$ is positive whenever the curve is situated above the number line, it is negative if the curve is found below the number line). Their union represents the solution of inequality.



Remark:

- (i) Points of discontinuity will never be included in the answer.
 (ii) If you are asked to find the intervals where $f(x)$ is non-negative or non-positive then make the intervals closed corresponding to the roots of the numerator and let it remain open corresponding to the roots of denominator.

¹ The point for which $f(x)$ vanishes (becomes zero) are called zeros of function e.g. $x = a$.

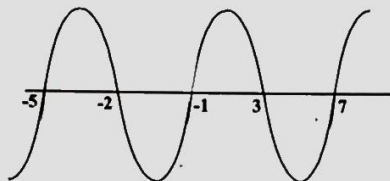
² The points $x = b_j$ are the point of the discontinuity of the function $f(x)$.

³ If the exponents of a factor is odd then the point is called a simple point.

⁴ If the exponent of a factor is even then the point is called a double point.

Illustration 3: Let $f(x) = \frac{(x-3)(x+2)(x+5)}{(x+1)(x-7)}$. Find intervals, where $f(x)$ is positive or negative.

Solution:



$$f(x) > 0 \quad \forall x \in (-5, -2) \cup (-1, 3) \cup (7, \infty)$$

$$\text{and } f(x) < 0 \quad \forall x \in (-\infty, -5) \cup (-2, -1) \cup (3, 7).$$

Illustration 4: Find the set of all x for which $\frac{2x}{2x^2 + 5x + 2} > \frac{1}{x+1}$.

Solution:

$$\begin{aligned} \text{We have } \frac{2x}{2x^2 + 5x + 2} - \frac{1}{x+1} &> 0 \Rightarrow \frac{3x+2}{(x+1)(2x+1)(x+2)} < 0 \\ &\Rightarrow \frac{(3x+2)^2}{(x+1)(2x+1)(x+2)(3x+2)} < 0 \\ &\Rightarrow \frac{(3x+2)^2}{2.3(x+\frac{1}{2})(x+\frac{2}{3})(x+1)(x+2)} < 0. \quad \dots(1) \end{aligned}$$

There are five intervals $x < -2$, $-2 < x < -1$, $-1 < x < -2/3$, $-2/3 < x < -1/2$, $x > -1/2$.

The inequality (1) will hold for $-2 < x < -1$ and for $-2/3 < x < -1/2$.

Hence $-2 < x < -1$ and $-2/3 < x < -1/2$.

SOLVED PROBLEMS

SUBJECTIVE

Problem 1: Let S denotes the set for all real values of the parameter 'a' for which every solution of the inequality $\log_{1/2} x^2 \geq \log_{1/2} (x+2)$ is a solution of the inequality $49x^2 - 4a^4 \leq 0$. Find S .

Solution:

$$\text{We have } \log_{1/2} x^2 \geq \log_{1/2} (x+2)$$

$$\Rightarrow x^2 \leq x+2$$

$$\Rightarrow -1 \leq x \leq 2$$

... (1)

$$\text{and } 49x^2 - 4a^4 \leq 0 \text{ i.e., } x^2 \leq \frac{4a^4}{49} \Rightarrow -\frac{2a^2}{7} \leq x \leq \frac{2a^2}{7} \quad \dots (2)$$

From (1) and (2) we have

$$-\frac{2a^2}{7} \leq -1 \text{ and } 2 \leq \frac{2a^2}{7}$$

$$\text{i.e., } a^2 \geq 7/2 \text{ and } a^2 \geq 7 \Rightarrow a^2 \geq 7$$

$$\Rightarrow a \in (-\infty, -\sqrt{7}] \cup [\sqrt{7}, \infty)$$

$$\text{Thus } S = (-\infty, -\sqrt{7}] \cup [\sqrt{7}, \infty).$$

Problem 2:

Let x, y, z be real variables satisfying the equations $x + y + z = 6$ and $xy + yz + zx = 7$. Then find the range in which the variables can lie.

Solution:

We are given that

$$x + y + z = 6 \quad \dots (1)$$

$$\text{and } xy + yz + zx = 7. \quad \dots (2)$$

$$\text{From (1), } z = 6 - x - y.$$

Putting the value of z in (2), we get

$$xy + y(6 - x - y) + x(6 - x - y) = 7 \text{ or, } y^2 + y(x - 6) + x^2 - 6x + 7 = 0.$$

$$\text{Since } y \text{ is real, } (x - 6)^2 - 4(x^2 - 6x + 7) \geq 0$$

$$\text{or, } 3x^2 - 12x - 8 \leq 0 \Rightarrow \frac{6 - 2\sqrt{15}}{3} \leq x \leq \frac{6 + 2\sqrt{15}}{3}.$$

Since (1) and (2) are symmetrical in x, y and z , all the variables lie in the interval

$$\left[\frac{6 - 2\sqrt{15}}{3}, \frac{6 + 2\sqrt{15}}{3} \right].$$

OBJECTIVE

- Problem 1:** If the expression $\left(mx - 1 + \frac{1}{x}\right)$ is non-negative for all positive real x , then the minimum value of m must be
- (A) $-1/2$ (B) 0
(C) $1/4$ (D) $1/2$

Solution:

We know that $ax^2 + bx + c \geq 0$ if $a > 0$ and $b^2 - 4ac \leq 0$.

$$\text{So, } mx - 1 + \frac{1}{x} \geq 0 \Rightarrow \frac{mx^2 - x + 1}{x} \geq 0$$

$$\Rightarrow mx^2 - x + 1 \geq 0 \text{ as } x > 0.$$

$$\text{Now, } mx^2 - x + 1 \geq 0 \text{ if } m > 0 \text{ and } 1 - 4m \leq 0$$

$$\text{or if } m > 0 \text{ and } m \geq 1/4.$$

Thus, the minimum value of m is $1/4$.

Hence (C) is the correct answer.

- Problem 2:** The largest negative integer which satisfies $\frac{x^2 - 1}{(x - 2)(x - 3)} > 0$ is
- (A) -4 (B) -3
(C) -1 (D) -2

Solution:

By wavy curve method

$$\frac{x^2 - 1}{(x-2)(x-3)} > 0 \Rightarrow x \in (-\infty, -1) \cup (1, 2) \cup (3, \infty).$$

Therefore largest negative integer is -2 .

Hence (D) is the correct answer.

Problem 3: The complete set of values of x satisfying $\log_{x^2}(x+1) > 0$ is

(A) $(1, \infty)$

(B) $(-1, 2) \sim \{0\}$

(C) $(-1, 1) \sim \{0\}$

(D) $(-1, 0) \cup (1, \infty)$

Solution: If $x^2 > 1$, then $x + 1 > 0 \Rightarrow x > 0$

$$\Rightarrow x \in (1, \infty)$$

If $0 < x^2 < 1$, then $0 < x + 1 < 1$

$$\Rightarrow x \in (-1, 0)$$

Hence (D) is the correct answer.

EXERCISE - 2

- Find the roots α, β, γ of $x^3 - 11x^2 + 36x - 36 = 0$ if $\frac{2}{\beta} = \frac{1}{\alpha} + \frac{1}{\gamma}$.
- If the roots of $x^3 + px^2 + qx + r = 0$ are in G.P., find the relation between p, q, r .
- In a triangle PQR; $\angle R = \frac{\pi}{2}$. If $\tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of the equation $ax^2 + bx + c = 0$, ($a \neq 0$), then show that $a + b - c = 0$.
- If α, β, γ are the roots of $x^3 - x^2 - 1 = 0$, then find the value of $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$.
- Prove that $x^6 + x^4 + x^2 + x + 3 = 0$ has no positive real roots.
- Solve the inequality $\frac{x-1}{x^2-4x+3} < 1$.
- Solve $\frac{(x-1)^2(3x+2)^{70}(x^2+x+1)^{31}}{(x-7)(3x-7)^{71}(x+3)x^{99}} \leq 0$.
- Find the solution set of $\frac{(x-1)(x-2)^2(x+4)}{(x+2)(x-3)} \geq 0$.
- Find the solution set of $\frac{|x-1|(x-3)(x-5)^{2010}}{(|x|-3)(|x|+1)} \geq 0$.
- The greatest negative integer satisfying $x^2 - 4x - 77 < 0$ and $x^2 > 4$ is
 (A) -5 (B) -6
 (C) -3 (D) none of these

11. The range of x for which $x > 1/x$, is

(A) $(1, \infty)$

(B) $(-1, 0)$

(C) $(-1, 0) \cup (1, \infty)$

(D) $(-\infty, -1)$

Quadratic Expression

The expression $ax^2 + bx + c$ is said to be a real quadratic expression in x where a, b, c are real and $a \neq 0$. Let $f(x) = ax^2 + bx + c$ where $a, b, c \in \mathbb{R}$ ($a \neq 0$).

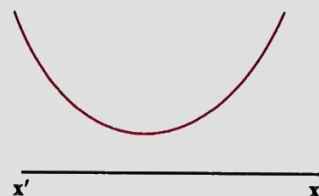
$f(x)$ can be rewritten as $f(x) = a \left\{ \left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right\} = a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a^2} \right]$, where $D = b^2 - 4ac$ is the discriminant of the quadratic expression. Then $y = f(x)$ represents a parabola whose axis is parallel to the y -axis, with vertex at $A \left(-\frac{b}{2a}, -\frac{D}{4a} \right)$.

Depending on the sign of a and $b^2 - 4ac$, $f(x)$ may be positive, negative or zero. This gives rise to the following cases:

- (i) $a > 0$ and $b^2 - 4ac < 0$

$$\Leftrightarrow f(x) > 0 \quad \forall x \in \mathbb{R}.$$

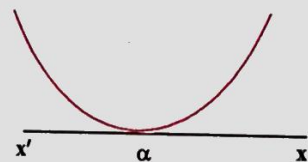
In this case the parabola always remains above the x -axis.



- (ii) $a > 0$ and $b^2 - 4ac = 0$

$$\Rightarrow f(x) \geq 0 \quad \forall x \in \mathbb{R}.$$

In this case the parabola touches the x -axis at one point and remains concave upwards.



- (iii) $a > 0$ and $b^2 - 4ac > 0$.

Let $f(x) = 0$ have two real roots α and β ($\alpha < \beta$).

Then $f(x) > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$,

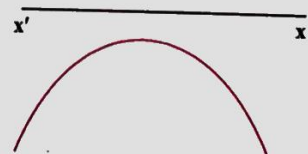
and $f(x) < 0 \quad \forall x \in (\alpha, \beta)$.



- (iv) $a < 0$ and $b^2 - 4ac < 0$

$$\Leftrightarrow f(x) < 0 \quad \forall x \in \mathbb{R}.$$

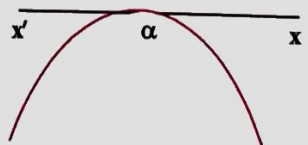
In this case the parabola always remains below the x -axis.



- (v) $a < 0$ and $b^2 - 4ac = 0$

$$\Leftrightarrow f(x) \leq 0 \quad \forall x \in \mathbb{R}.$$

In this case the parabola touches the x -axis and lies below the x -axis.



- (vi) $a < 0$ and $b^2 - 4ac > 0$

Let $f(x) = 0$ have two real roots α and β ($\alpha < \beta$).

Then $f(x) < 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$

and $f(x) > 0 \quad \forall x \in (\alpha, \beta)$.

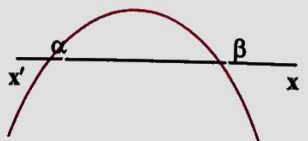


Illustration 1: Let $f(x)$ be a quadratic expression which is positive $\forall x \in \mathbb{R}$. If $g(x) = f(x) + f'(x) + f''(x)$, then prove that $g(x) > 0 \forall x \in \mathbb{R}$.

Solution: Let $f(x) = ax^2 + bx + c$. Then $g(x) = f(x) + f'(x) + f''(x)$
 $= ax^2 + bx + c + (2ax + b) + 2a = ax^2 + (b + 2a)x + c + b + 2a$.
 Given $f(x) = ax^2 + bx + c > 0 \forall x \in \mathbb{R}$
 $\Rightarrow b^2 - 4ac < 0$ and $a > 0$ (1)
 Now discriminant of equation $g(x) = 0$ is
 $D = (b + 2a)^2 - 4a(c + b + 2a)$
 $= b^2 + 4a^2 + 4ab - 4ac - 4ab - 8a^2$
 $= b^2 - 4a^2 - 4ac = (b^2 - 4ac) - 4a^2 < 0$ and $a > 0$ from (1)
 $\Rightarrow g(x) > 0 \forall x \in \mathbb{R}$.

Illustration 2: A real number 'a' is called a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq a$ is satisfied for all real x. Find the set of all good real numbers.

Solution: We have $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq a \forall x \in \mathbb{R}$
 $\Rightarrow 2x^2 + 2x + 3 \leq a(x^2 + x + 1) \forall x \in \mathbb{R}$
 $\Rightarrow (2 - a)x^2 + (2 - a)x + (3 - a) \leq 0 \forall x \in \mathbb{R}$
 $2 - a < 0$ and $(2 - a)^2 - 4(2 - a)(3 - a) \leq 0 \forall x \in \mathbb{R}$
 $\Rightarrow a > 2$ and $a \leq 2$ or $a \geq \frac{10}{3} \Rightarrow a \geq \frac{10}{3} \therefore a \in \left[\frac{10}{3}, \infty\right)$.

Illustration 3: If $ax^2 - bx + 5 = 0$ does not have 2 distinct real roots, then find the minimum value of $5a + b$.

Solution: Let $f(x) = ax^2 - bx + 5$. Since $f(x) = 0$ does not have 2 distinct real roots, we have either $f(x) \geq 0 \forall x \in \mathbb{R}$ or $f(x) \leq 0 \forall x \in \mathbb{R}$.
 But $f(0) = 5 > 0$, so $f(x) \geq 0 \forall x \in \mathbb{R}$. In particular $f(-5) \geq 0$
 $\Rightarrow 25a + 5b + 5 \geq 0 \Rightarrow 5a + b \geq -1$.
 Hence the least value of $5a + b$ is -1 .

SOLVED PROBLEMS

SUBJECTIVE

Problem 1: Find the range in which the value of the function $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ lies for all real values of x.

Solution: Let $\frac{x^2 + 34x - 71}{x^2 + 2x - 7} = y$
 $\Rightarrow x^2(y - 1) - 2x(17 - y) + 71 - 7y = 0$.
 For real values of x, $(17 - y)^2 - (y - 1)(71 - 7y) \geq 0$
 $\Rightarrow 8y^2 - 112y + 360 \geq 0 \Rightarrow y^2 - 14y + 45 \geq 0$

$$\Rightarrow (y-9)(y-5) \geq 0 \Rightarrow y \leq 5 \text{ or } y \geq 9.$$

Thus for real values of x , the given expression is either ≤ 5 or ≥ 9 .

Problem 2: If $\frac{x^2 + ax + 3}{x^2 + x + a}$ takes all real values for possible real values of x then prove that $4a^3 + 39 < 0$.

Solution: Let $\frac{x^2 + ax + 3}{x^2 + x + a} = y \Rightarrow x^2(1-y) - x(y-a) + 3 - ay = 0$.

$$\text{For } x \text{ to be real, } (y-a)^2 - 4(1-y)(3-ay) \geq 0$$

$$\text{or } y^2 + a^2 - 2ay - 4(3-ay-3y+ay^2) \geq 0 \text{ for all real } y$$

$$\text{or } y^2(1-4a) + y(4a+12-2a) + a^2 - 12 \geq 0 \text{ for all real } y.$$

For above expression to be true for all values of y , $1-4a > 0$ i.e. $a < 1/4$ and

$$4(a+6)^2 - 4(a^2 - 12)(1-4a) \leq 0$$

$$\text{i.e. } (a+6)^2 - (a^2 - 4a^3 - 12 + 48a) \leq 0$$

$$\Rightarrow a^2 + 12a + 36 - a^2 + 4a^3 + 12 - 48a \leq 0$$

$$\text{or, } 4a^3 - 36a + 48 \leq 0 \text{ or, } 4a^3 \leq 36a - 48$$

$$\text{or, } 4a^3 < -39 \text{ (as } a < 1/4) \Rightarrow 4a^3 + 39 < 0.$$

Problem 3: If $(y^2 - 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in \mathbb{R}$, then Show that the interval in which y lies is $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$.

Solution: Here $(y^2 - 5y + 3)(x^2 + x + 1) < 2x \quad \forall x \in \mathbb{R}$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1}.$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = p \Rightarrow px^2 + (p-2)x + p = 0.$$

$$\text{Since } x \text{ is real, } (p-2)^2 - 4p^2 \geq 0 \Rightarrow -2 \leq p \leq 2/3$$

$$\Rightarrow \text{Minimum value of } \frac{2x}{x^2 + x + 1} \text{ is } -2,$$

$$\text{so that, } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0$$

$$\Rightarrow y \in \left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right).$$

OBJECTIVE

Problem 1: If the inequality $\frac{mx^2 + 3x + 4}{x^2 + 2x + 2} < 5$ is satisfied for all $x \in \mathbb{R}$, then

(A) $1 < m < 5$

(B) $-1 < m < 5$

(C) $1 < m < 6$

(D) $m < \frac{71}{24}$

Solution: We have $x^2 + 2x + 2 = (x+1)^2 + 1 > 0, \forall x \in \mathbb{R}$.

$$\text{Therefore, } \frac{mx^2 + 3x + 4}{x^2 + 2x + 2} < 5 \Rightarrow mx^2 + 3x + 4 < 5(x^2 + 2x + 2)$$

$$\Rightarrow (m-5)x^2 - 7x - 6 < 0, \forall x \in \mathbb{R}.$$

This is possible if $D = b^2 - 4ac = 49 + 24(m-5) < 0$ and $m-5 < 0$

$$\Rightarrow m < \frac{71}{24}.$$

Hence (D) is the correct answer.

EXERCISE – 3

1. If the equation $ax^2 + bx + c = 0$ does not have 2 distinct real roots and $a + c > b$, then prove that $f(x) \geq 0, \forall x \in \mathbb{R}$.
2. Find the least value of n such that $(n-2)x^2 + 8x + n + 4 > 0, \forall x \in \mathbb{R}$ where $n \in \mathbb{N}$.
3. Let $y = \frac{x^2 + 3x + 1}{x^2 + x + 1}, \forall x \in \mathbb{R}$. Find the range of y .
4. Show that if x is real, the expression $\frac{x^2 - bc}{2x - b - c}$ has no value between b and c .
5. If x is real, then the least value of the expression $\frac{x^2 - 6x + 5}{x^2 + 2x + 1}$ is
 (A) -1 (B) -1/2
 (C) -1/3 (D) none of these
6. The condition that $x^4 + ax^3 + bx^2 + cx + d$ is a perfect square, is
 (A) $c^2 = ad$ (B) $c^2 = ad^2$
 (C) $c^2 = a^2d^2$ (D) $c^2 = a^2d$

Interval in which the Roots Lie

In some problems we want the roots of the equation $ax^2 + bx + c = 0$ to lie in a given interval. For this we impose conditions on a, b and c . Let $f(x) = ax^2 + bx + c$.

- (i) If both the roots are positive i.e. they lie in $(0, \infty)$, then the sum of the roots as well as the product of the roots must be positive.

$$\Rightarrow \alpha + \beta = -\frac{b}{a} > 0 \text{ and } \alpha\beta = \frac{c}{a} > 0 \text{ with } b^2 - 4ac \geq 0.$$

Similarly, if both the roots are negative i.e. they lie in $(-\infty, 0)$ then the sum of the roots will be negative and the product of the roots must be positive.

$$\text{i.e. } \alpha + \beta = -\frac{b}{a} < 0 \text{ and } \alpha\beta = \frac{c}{a} > 0 \text{ with } b^2 - 4ac \geq 0.$$

- (ii) Both the roots are greater than a given number k if the following three conditions are satisfied
 $D \geq 0, -\frac{b}{2a} > k$ and $a.f(k) > 0$.
- (iii) Both the roots will be less than a given number k if the following conditions are satisfied:
 $D \geq 0, -\frac{b}{2a} < k$ and $a.f(k) > 0$.
- (iv) Both the roots will lie in the given interval (k_1, k_2) if the following conditions are satisfied $D \geq 0$,
 $k_1 < -\frac{b}{2a} < k_2$ and $a.f(k_1) > 0, a.f(k_2) > 0$.

- (v) Exactly one of the roots lies in the given interval (k_1, k_2) if $f(k_1) \cdot f(k_2) < 0$.
 (vi) A given number k will lie between the roots if $a \cdot f(k) < 0$.
 In particular, the roots of the equation will be of opposite signs if 0 lies between the roots
 $\Rightarrow a \cdot f(0) < 0$.

Illustration 1: Find the values of the parameter a for which the roots of the quadratic equation $x^2 + 2(a-1)x + a + 5 = 0$ are

- | | |
|---|------------------------|
| (i) real and distinct, | (ii) equal, |
| (iii) not real, | (iv) opposite in sign, |
| (v) equal in magnitude but opposite in sign, | |
| (vi) positive, | (vii) negative, |
| (viii) such that one root is greater than 3, and the other is smaller than 3, | |
| (ix) greater than 3, | (x) smaller than 3, |
| (xi) such that exactly one root lies in the interval $(1, 3)$, | |
| (xii) such that both the roots lie in the interval $(1, 3)$, | |
| (xiii) such that one root is greater than 3 and the other root is smaller than 1. | |

Solution:

Let $f(x) = x^2 + 2(a-1)x + a + 5 = Ax^2 + Bx + C$ (say)

$\Rightarrow A = 1, B = 2(a-1), C = a + 5$.

Also $D = B^2 - 4AC = 4(a-1)^2 - 4(a+5) = 4(a+1)(a-4)$.

- (i) $D > 0 \Rightarrow (a+1)(a-4) > 0$
 $\Rightarrow a \in (-\infty, -1) \cup (4, \infty)$.
- (ii) $D = 0 \Rightarrow (a+1)(a-4) = 0 \Rightarrow a = -1, 4$.
- (iii) $D < 0 \Rightarrow (a+1)(a-4) < 0$
 $\Rightarrow a \in (-1, 4)$.
- (iv) This means that 0 lies between the roots of the given equation.
 $\Rightarrow f(0) < 0$ and $D > 0$ i.e. $a \in (-\infty, -1) \cup (4, \infty)$
 and $a + 5 < 0 \Rightarrow a < -5$
 or $a \in (-\infty, -5)$.
- (v) This means that the sum of the roots is zero
 $\Rightarrow -2(a-1) = 0$ and $D > 0$ i.e. $a \in (-\infty, -1) \cup (4, \infty)$
 and $a = 1$
 which does not belong to $(-\infty, -1) \cup (4, \infty) \Rightarrow a$ is null set ϕ .
- (vi) This implies that both the roots are greater than zero
 $\Rightarrow -B/A > 0, C/A > 0, D \geq 0$
 $\Rightarrow -(a-1) > 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$
 $\Rightarrow a < 1, -5 < a, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-5, -1]$.
- (vii) This implies that both the roots are less than zero
 $\Rightarrow -B/A < 0, C/A > 0, D \geq 0$
 $\Rightarrow -(a-1) < 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$
 $\Rightarrow a > 1, a > -5, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in [4, \infty)$.
- (viii) This means that 3 lies between the roots
 $\Rightarrow f(3) < 0$ and $D > 0$
 $\Rightarrow 9 + 2(a-1)(3) + a + 5 < 0$ and $a \in (-\infty, -1) \cup (4, \infty)$
 $\Rightarrow a < -8/7$ and $a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a \in \left(-\infty, -\frac{8}{7}\right)$.
- (ix) In this case
 $-\frac{B}{2A} > 3, A \cdot f(3) > 0$ and $D \geq 0$
 $\Rightarrow -(a-1) > 3, 7a + 8 > 0$ and $a \in (-\infty, -1) \cup [4, \infty)$

$$\Rightarrow a < -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty).$$

Since no value of a can satisfy these conditions simultaneously, there can be no value of a for which both the roots will be greater than 3.

(x) In this case

$$-\frac{B}{2A} < 3, A.f(3) > 0 \text{ and } D \geq 0$$

$$\Rightarrow a > -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a \in \left(-\frac{8}{7}, -1\right] \cup [4, \infty).$$

(xi) In this case $f(1).f(3) < 0$ and $D \geq 0$

$$\Rightarrow (3a + 4)(7a + 8) < 0 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a \in \left(-\frac{4}{3}, -\frac{8}{7}\right).$$

(xii) In this case $1 < -\frac{B}{2A} < 3, A.f(1) > 0, A.f(3) > 0, D \geq 0$

$$\Rightarrow 1 < -1(a - 1) < 3, 3a + 4 > 0, 7a + 8 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow -2 < a < 0, a > -4/3, a > -8/7, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a \in \left(-\frac{8}{7}, -1\right].$$

(xiii) In this case $D > 0$ and $f(1) < 0$ i.e. $3a + 4 < 0$

$$\Rightarrow a < -4/3 \text{ and } f(3) < 0 \text{ i.e. } 7a + 8 < 0$$

$$\Rightarrow a < -8/7 \text{ and } a \in (-\infty, -1] \cup (4, \infty).$$

Combining we get, $a \in (-\infty, -4/3).$

Illustration 2: If α is a root of the equation $ax^2 + bx + c = 0$ and β is a root of the equation $-ax^2 + bx + c = 0$, then prove that there will be a root of the equation $\frac{a}{2}x^2 + bx + c = 0$ lying between α and β .

Solution:

$$\text{Let } f(x) = \frac{a}{2}x^2 + bx + c$$

$$\Rightarrow f(\alpha) = \frac{a}{2}\alpha^2 + b\alpha + c = a\alpha^2 + b\alpha + c - \frac{a}{2}\alpha^2$$

$$= -\frac{a}{2}\alpha^2 \quad (\text{As } \alpha \text{ is a root of } ax^2 + bx + c = 0).$$

$$\text{Also } f(\beta) = \frac{a}{2}\beta^2 + b\beta + c = -a\beta^2 + b\beta + c + \frac{3}{2}a\beta^2$$

$$= \frac{3}{2}a\beta^2 \quad (\text{As } \beta \text{ is a root of } -ax^2 + bx + c = 0).$$

$$\text{Now } f(\alpha).f(\beta) = \frac{-3}{4}a^2\alpha^2\beta^2 < 0 \Rightarrow f(x) = 0 \text{ has one real root between } \alpha \text{ and } \beta.$$

Illustration 3: Find the values of 'a' for which $4^t - (a - 4)2^t + \frac{9}{4}a < 0 \quad \forall t \in (1, 2).$

Solution:

$$\text{Let } 2^t = x \text{ and } f(x) = x^2 - (a - 4)x + \frac{9}{4}a.$$

$$\text{We want } f(x) < 0 \quad \forall x \in (2^1, 2^2) \text{ i.e. } \forall x \in (2, 4)$$

(i) Since coefficient of x^2 in $f(x)$ is positive, $f(x) < 0$ for some x only when roots of $f(x) = 0$ are real and distinct

$$\Rightarrow D > 0 \Rightarrow a^2 - 17a + 16 > 0 \Rightarrow a < 1 \text{ or } a > 16. \quad \dots (1)$$

(ii) Since we want $f(x) < 0 \forall x \in (2, 4)$, one of the roots of $f(x)$ should be less than 2 and the other must be greater than 4 i.e. $f(2) \leq 0$ and $f(4) \leq 0$. $a \leq -48$ and $a \geq 128/7$, which is not possible.

Hence no such 'a' exist.

SOLVED PROBLEMS

SUBJECTIVE

Problem 1: The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + \beta x + \gamma = 0$ are in A.P. Find the intervals in which β and γ lie.

Solution:

Given that $x_2 - x_1 = x_3 - x_2$

$$\text{and } x^3 - x^2 + \beta x + \gamma = 0. \quad \dots(1)$$

x_1, x_2 and x_3 are roots of equation (1), then

$$x_1 + x_2 + x_3 = 1 = 3x_2 \Rightarrow x_2 = 1/3 \text{ and } x_1 + x_3 = 2/3.$$

$$\text{Also } x_1x_2 + x_2x_3 + x_1x_3 = \beta \text{ i.e. } \frac{x_1}{3} + \frac{x_3}{3} + x_1x_3 = \beta. \quad \dots(2)$$

$$\text{Again } x_1x_2x_3 = -\gamma \Rightarrow \frac{x_1x_3}{3} = -\gamma \Rightarrow x_1x_3 = -3\gamma \quad \dots(3)$$

$$\Rightarrow x_1 \left(\frac{2}{3} - x_1 \right) = -3\gamma \Rightarrow 3x_1^2 - 2x_1 - 9\gamma = 0.$$

$$\text{For real value of } x_1, (-2)^2 + 108\gamma \geq 0 \Rightarrow \gamma \geq -\frac{1}{27}.$$

From equation (2), we have

$$x_1 + x_3 + 3x_1x_3 = 3\beta \Rightarrow \beta = \frac{2}{9} - 3\gamma.$$

$$\text{Since } \gamma \geq -\frac{1}{27}, \beta \leq \frac{2}{9} + \frac{3}{27} \Rightarrow \beta \leq \frac{1}{3}.$$

Problem 2: For what value of θ , 1 lies between the roots of the quadratic equation $3x^2 - 3 \sin \theta x - 2 \cos^2 \theta = 0$

Solution:

$$\text{Let } f(x) = 3x^2 - 3 \sin \theta x - 2 \cos^2 \theta.$$

$$\text{The coefficient of } x^2 > 0 \Rightarrow f(1) < 0$$

$$\Rightarrow 3 - 3 \sin \theta - 2 \cos^2 \theta < 0 \Rightarrow 2 \sin^2 \theta - 3 \sin \theta + 1 < 0$$

$$\Rightarrow (2 \sin \theta - 1)(\sin \theta - 1) < 0$$

$$\Rightarrow 1/2 < \sin \theta < 1 \Rightarrow 2n\pi + \pi/6 < \theta < 2n\pi + 5\pi/6.$$

Problem 3: Find the value of 'a' for which $ax^2 + (a-3)x + 1 < 0$ for at least one positive real x.

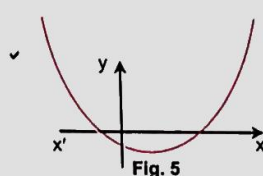
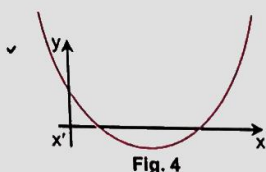
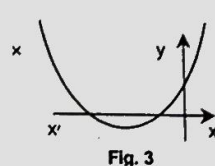
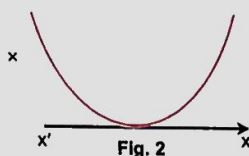
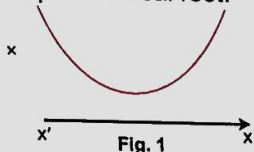
Solution:

$$\text{Let } f(x) = ax^2 + (a-3)x + 1$$

Case (i) If $a < 0$, then $f(x)$ will be positive only for those values of x which lie between the roots of the equation $f(x) = 0$.

As the interval between the roots can not cover all the positive real numbers implies $f(x) < 0$ for at least one positive real x

Case(ii) If $a > 0$, then $f(x)$ will be negative only for those values of x which lie between the roots of $f(x) = 0$. From the graphs we can see that $f(x)$ will be less than zero for at least one positive real x , when $f(x) = 0$ has distinct roots and at least one of these roots is positive real root.



For this $D > 0$, i.e. $(a-3)^2 - 4a > 0 \Rightarrow a < 1$ or $a > 9$ (1)

Both the roots are non-positive $\Rightarrow$ sum ≤ 0 and product ≥ 0

$\Rightarrow a \geq 3$

... (2)

$\Rightarrow$ at least one roots is positive if $a < 3$, and (1) is satisfied.

Combining (1) and (2), we get $a < 1$ so that, $0 < a < 1$.

Case (iii) If $a = 0$, the given inequality is satisfied for at least one $x > 0$.

Hence the required set of values of a is $(-\infty, 1)$.

OBJECTIVE

Problem 1: The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is

(A) $(-\infty, 0)$

(B) $(0, 1)$

(C) $(1, \infty)$

(D) $(0, \infty)$

Solution: Since the roots of the given equation are of opposite sign, product of the roots < 0

$$\Rightarrow \frac{p(p-1)}{3} < 0 \Rightarrow p(p-1) < 0 \Rightarrow 0 < p < 1.$$

$$\text{For real and distinct roots } \frac{1}{2} - \frac{\sqrt{21}}{6} < p < \frac{1}{2} + \frac{\sqrt{21}}{6} \Rightarrow 0 < p < 1.$$

Hence (B) is the correct answer.

Problem 2: If $c > 0$ and $4a + c < 2b$ then $ax^2 - bx + c = 0$ has a root in the interval

(A) $(0, 2)$

(B) $(2, 4)$

(C) $(0, 1)$

(D) $(-2, 0)$

Solution: Let $f(x) = ax^2 - bx + c$

$f(0) = c > 0$ and $f(2) = 4a - 2b + c < 0$ so that $f(x) = 0$ has a root in the interval $(0, 2)$.

Hence (A) is the correct answer.

EXERCISE – 4

1. For what values of $a \in \mathbb{R}$, the quadratic equation $(a^2 + 1)x^2 - (a + 1)x + (a^2 - a - 2) = 0$ will have roots of opposite sign.
2. For what values of a , does the equation $ax^2 - (a + 1)x + 3 = 0$, have roots lying between 1 and 2.
3. If the equation $x^2 + ax - a^2 - 1 = 0$ has roots of opposite signs, then determine the interval in which 'a' lies.
4. If both the roots of $x^2 - ax + a = 0$ are greater than 2, then determine the interval in which 'a' lies.
5. Let $x^2 - (m - 3)x + m = 0$ ($m \in \mathbb{R}$) be a quadratic equation. Find the value of m for which the roots

(i) are real and distinct,	(ii) are equal,
(iii) are not real,	(iv) are opposite in sign,
(v) are equal in magnitude but opposite in sign,	(vi) are positive,
(vii) are negative,	(viii) are such that at least one is positive.
6. The equation $ax^2 + bx + c = 0$ has real and positive roots. Prove that the roots of the equation $a^2x^2 + a(3b - 2c)x + (2b - c)(b - c) + ac = 0$ are real and positive.
7. If the equation $(a - 5)x^2 + 2(a - 10)x + a + 10 = 0$ has roots of the opposite sign, then

(A) $a > 10$	(B) $-15 < a < 5$
(C) $-10 < a < 5$	(D) none of these
8. The set of values of 'a' for which 1 lies between the roots of $x^2 - ax - a + 3 = 0$ is

(A) $(-\infty, -6)$	(B) $(-\infty, +6)$
(C) $(-\infty, -6) \cup (2, \infty)$	(D) $(2, \infty)$

MISCELLANEOUS EXERCISE

1. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, show that $p + q = 2r$ and the product of the roots is $-\frac{1}{2}(p^2 + q^2)$.
2. One of the roots of the equation $x^2 + px + q = 0$ ($p, q \in \mathbb{R}$) is $a + ib$ ($b \neq 0$). Show that $q > 0$ and also $a < 0$ if and only if $p > 0$.
3. The coefficient of x in the quadratic equation $x^2 + px + q = 0$ was taken as 17 in place of 13. Its roots were found to be -2 , and -15 . Find the roots of the original equation.
4. Prove that the arithmetic mean of the roots of $x^2 - 2ax + b^2 = 0$ is equal to the geometric mean of the roots of the equation $x^2 - 2bx + a^2 = 0$.
5. If the ratio of the roots of the quadratic equation $x^2 + px + q = 0$ be equal to the ratio of the roots of $x^2 + lx + m = 0$, then prove that $p^2m = l^2q$.